# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

**Full Syllabus (Chapters 1–14)** | Class 10 Maths | Maximum Marks: 50

### MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 2310 | Highest powers of $2 \times 3 \times 5$, $2 \times 3 \times 7$, $2 \times 3 \times 11$ |
| 2 | **(c)** 28 | $(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$ |
| 3 | **(d)** -4 | $k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines |
| 4 | **(b)** 29 | $(\alpha+\beta)^2 - 2\alpha\beta$ |
| 5 | **(a)** 0 | Common difference is $-1$; $a_{p+q} = 0$ |
| 6 | **(c)** 8 | $(m+n)^2 - m^2$ |
| 7 | **(c)** 12 | Let the point be $(0, y)$ |
| 8 | **(d)** $\frac{25}{9}$ | $1 + \tan^2\theta$ |
| 9 | **(c)** $2+2\sqrt{3}$ m | $h(\cot 30^\circ - \cot 45^\circ) = 4$ gives $h = 4 \div (\sqrt{3}-1)$; Height $= 2+2\sqrt{3}$ m (distances $6+2\sqrt{3}$ m and $2+2\sqrt{3}$ m) |
| 10 | **(d)** 48 cm | $AM = \frac{30\times40}{50} = 24$ cm; $AB = 2AM = 48$ cm |
| 11 | **(c)** 4.47 cm² | Sector $= 25.67$ cm²; triangle $= \frac{\sqrt{3}}{4}r^2 = 21.19$ cm²; Segment $= 25.67 - 21.19 = 4.47$ cm² |
| 12 | **(d)** 4410 cm³ | Sphere $= \frac{4}{3}\pi\left(\frac{21}{2}\right)^3 = 4851$ cm³; cube $= 9261$ cm³; Wasted $= 4410$ cm³ |
| 13 | **(b)** 23 | $\frac{4325+25p}{77+p} = 49$ gives $(25-49)p = -552$; $p = 23$ |
| 14 | **(d)** $\frac{5}{36}$ | Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$ |
| 15 | **(d)** 60 | LCM of 1 to 6 |
| 16 | **(b)** 30 | $\alpha\beta(\alpha+\beta)$ |
| 17 | **(d)** 2 | Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$ |
| 18 | **(c)** 24 | $(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$ |
| 19 | **(c)** 54 | $a_n = \frac{2S_n}{n} - a$ |
| 20 | **(a)** 6 | Find $b$ by Pythagoras |
| 21 | **(c)** 63 | Twice the area |
| 22 | **(b)** $-\frac{7}{25}$ | $\cos^2\theta - \sin^2\theta$ |
| 23 | **(c)** 20 m | Height of the top of the flagstaff $= d\tan 60^\circ = 30$ m; Length $= 30 - 10 = 20$ m |
| 24 | **(b)** 14 cm | Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2 \times 7 = 14$ cm |
| 25 | **(b)** 150 mm | Circumference $= \pi d = 66$ mm; diameters $= 4 \times 21 = 84$ mm; Total $= 150$ mm |
| 26 | **(c)** 2425.5 cm³ | Cylinder $= \pi r^2(2r) = 7276.5$ cm³; sphere $= \frac{4}{3}\pi r^3 = 4851$ cm³; Unoccupied $= 2425.5$ cm³ |
| 27 | **(b)** 9 | $238.4 = 230 + \frac{50-(20+x)}{25} \times 10$ gives $x = 9$; $x = 9$; then $y = 100 - 82 - x = 9$ |
| 28 | **(b)** $\frac{1}{6}$ | Favourable outcomes: (1,1), (1,2), (1,3), (2,1), (2,2), (3,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$ |
| 29 | **(b)** 9 | HCF of the differences from the remainder |
| 30 | **(b)** 7 | $\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$ |
| 31 | **(b)** 8 | Solve for $x$ and $y$ |
| 32 | **(a)** 6 | $D = 0$ gives $k^2 = 4m$ |
| 33 | **(a)** 46 | $a_n = S_n - S_{n-1}$ |
| 34 | **(b)** $\frac{12}{5}$ | $\frac{1}{2}ab = \frac{1}{2}ch$ |
| 35 | **(c)** $\frac{16}{3}$ | Equate squared distances |
| 36 | **(a)** $\frac{24}{25}$ | $2 \cdot \frac{p}{c} \cdot \frac{q}{c}$ |
| 37 | **(c)** 2 minutes | Distance covered in 4 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it; Time left $= 4 \div 2 = 2$ minutes |

| | | |
|---|---|---|
| **38** | **(a)** 49 cm | $15x + 4 = 8x + 25$ gives $x = 3$; $QA = 15(3) + 4 = 49$ cm |
| **39** | **(c)** 1925 m² | Area $= \frac{180}{360} \times \frac{22}{7} \times 35^2$; Area $= 1925$ m$^2$ |
| **40** | **(b)** 15 cm | $H = 3h = 3 \times 5$; $H = 15$ cm |
| **41** | **(a)** Both A and R are true and R is the correct explanation of A. | HCF x LCM = 6 x 36 = 216 = 12 x 18, and R is the general result used to find the LCM. |
| **42** | **(a)** Both A and R are true and R is the correct explanation of A. | Each point where the graph meets the x-axis has $y = 0$, i.e. $p(x) = 0$; so R explains A. |
| **43** | **(a)** Both A and R are true and R is the correct explanation of A. | $3 + 1 = 4$ and $3 - 1 = 2$; the common point satisfies both equations, which is exactly the solution (R). |
| **44** | **(a)** Both A and R are true and R is the correct explanation of A. | $D = 9 - 40 = -31 < 0$, so there are no real roots, as stated in R. |
| **45** | **(a)** Both A and R are true and R is the correct explanation of A. | $7 - 3 = 11 - 7 = 15 - 11 = 4$; R is the defining property of an AP. |
| **46** | **(a)** Both A and R are true and R is the correct explanation of A. | $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{2}{3} = \frac{4}{EC} \Rightarrow EC = 6$; R is the Basic Proportionality Theorem. |
| **47** | **(a)** Both A and R are true and R is the correct explanation of A. | $\sqrt{9 + 16} = 5$; R is the formula used. |
| **48** | **(a)** Both A and R are true and R is the correct explanation of A. | $\frac{3}{4} + \frac{1}{4} = 1$; A is a particular case of the identity in R. |
| **49** | **(a)** Both A and R are true and R is the correct explanation of A. | With h fixed, smaller d gives larger $\tan\theta$, hence larger $\theta$; R is the explanation. |
| **50** | **(a)** Both A and R are true and R is the correct explanation of A. | $\angle OPT = 90^\circ$, so $OT^2 = 5^2 + 12^2 = 169$ and $OT = 13$ cm; R is what makes the triangle right-angled. |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

**Full Syllabus (Chapters 1–14)** | Class 10 Maths | Maximum Marks: 50

### MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 12 m | Both objects are on the same side, so the distances are subtracted; Distance $= 12$ m |
| 2 | **(b)** 120 cm | Half-chord $= \sqrt{68^2 - 32^2} = 60$ cm; Chord $= 2 \times 60 = 120$ cm |
| 3 | **(b)** 117 cm | Arc $= \frac{45}{360} \times 2\pi \times 42 = 33$ cm; Perimeter $= 33 + 84 = 117$ cm |
| 4 | **(b)** 14 cm | Cancel $\frac{1}{3}\pi r^2$: $h = 2r$; $h = 14$ cm |
| 5 | **(b)** 108 | Median class 100–120: $l = 100$, $cf = 44$, $f = 15$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 100 + \frac{50-44}{15} \times 20 = 108$ |
| 6 | **(c)** 5 | Cross-multiplying: $7(7 + x) = 4(16 + x)$; $x = 5$ |
| 7 | **(c)** $\frac{\sqrt{12}}{\sqrt{3}}$ | $\sqrt{12}/\sqrt{3} = 2$ |
| 8 | **(b)** 11 | $\frac{c}{a}$ for the monic cubic |
| 9 | **(a)** 29 | $(x + y)^2 - 2xy = 49 - 20$ |
| 10 | **(c)** $k \leq 3$ and $k \neq 0$ | $D = 36 - 12k \geq 0$ |
| 11 | **(a)** 1325 | Find $n$ first |
| 12 | **(c)** $\frac{9}{5}$ | $AB^2 = AD \cdot AC$ |
| 13 | **(d)** 50 | $s^2 = d^2/2$ |
| 14 | **(d)** 1 | Each pair sums to 1 |
| 15 | **(a)** 4 m | Part of the tower above the top of the building $= d \tan 30^\circ = 1$ m; Tower $= 4$ m |
| 16 | **(b)** 15 cm | $AF = AB - r = 45 - r$, $CD = BC - r = 60 - r$ and $AF + CD = AC$ (tangent lengths); $(45 - r) + (60 - r) = 75$ gives $r = 15$ cm |
| 17 | **(c)** $30^\circ$ | $\theta = \frac{22 \times 360}{2 \times \frac{22}{7} \times 42}$; $\theta = 30^\circ$ |
| 18 | **(c)** 4 : 9 | $2^2 : 3^2 = 4 : 9$; Ratio $= 4 : 9$ |
| 19 | **(c)** 63 | $l = 60$, $f_1 = 24$, $f_0 = 21$, $f_2 = 17$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{3}{10} \times 10 = 63$ |
| 20 | **(c)** $\frac{4}{5}$ | Favourable outcomes $=$ number of red balls; $P = \frac{20}{25} = \frac{4}{5}$ |
| 21 | **(a)** $2^3 \times 5^2 \times 3$ | Zeros at the end = min(power of 2, power of 5) |
| 22 | **(b)** 21 | $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$ |
| 23 | **(c)** 3 | Subtract the equations |
| 24 | **(b)** $\frac{5}{6}$ | $\frac{\alpha + \beta}{\alpha\beta}$ |
| 25 | **(b)** 11 | $S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$ |
| 26 | **(a)** 25 | Side $= s$ |
| 27 | **(d)** 28 | Use the area formula |
| 28 | **(a)** 1 | Pair as $\tan\theta \tan(90^\circ - \theta) = 1$ |
| 29 | **(d)** $9 + 3\sqrt{3}$ m | The points are on opposite sides, so the distances add; Distance $= 9 + 3\sqrt{3}$ m |
| 30 | **(a)** 36 cm | $PT^2 = OP^2 - OT^2 = 85^2 - 77^2 = 1296$; $PT = 36$ cm |
| 31 | **(d)** 1400 cm² | Circle $= 3850$ cm$^2$; square $= 2450$ cm$^2$; Difference $= 1400$ cm$^2$ |
| 32 | **(a)** 8 : 27 | $2^3 : 3^3 = 8 : 27$; Ratio $= 8 : 27$ |
| 33 | **(c)** 25 | $20 + \frac{25-18}{14} \times 10 = 25$ |
| 34 | **(d)** $\frac{19}{70}$ | Favourable numbers: 19 outcomes; number $= 19$; $P = \frac{19}{70}$ |
| 35 | **(a)** 144 | Highest powers of $2^2 \times 3^2$, $2^4 \times 3$, $2^3 \times 3^2$ |

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| **36** | **(b)** 7 | Sum of zeroes $= 9$, so other zero $= 9 - 2$ |
| **37** | **(b)** -6 | $k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines |
| **38** | **(a)** 53 | $(\alpha + \beta)^2 - 2\alpha\beta$ |
| **39** | **(b)** 0 | Common difference is $-1$; $a_{p+q} = 0$ |
| **40** | **(b)** 21 | $(m + n)^2 - m^2$ |
| **41** | **(a)** Both A and R are true and R is the correct explanation of A. | $\frac{1}{4} \times \frac{22}{7} \times 49 = 38.5$; R is the formula used. |
| **42** | **(a)** Both A and R are true and R is the correct explanation of A. | $2\pi r^2 + \pi r^2 = 3\pi r^2$; R gives the reason. |
| **43** | **(a)** Both A and R are true and R is the correct explanation of A. | $\frac{10+20}{2} = 15$; R is the definition used. |
| **44** | **(a)** Both A and R are true and R is the correct explanation of A. | No face of a die shows more than 6, so the event is impossible and its probability is 0. |
| **45** | **(a)** Both A and R are true and R is the correct explanation of A. | 5 is rational and $2\sqrt{3}$ is irrational, so the difference is irrational; R explains it. |
| **46** | **(c)** A is true but R is false. | Sum $= -\frac{b}{a} = \frac{8}{2} = 4$, so A is true; R is false because $\frac{c}{a}$ is the product. |
| **47** | **(c)** A is true but R is false. | $\frac{1}{2} = \frac{2}{4} \neq \frac{4}{12}$, so the lines are parallel and there is no solution: A true; R is false. |
| **48** | **(c)** A is true but R is false. | $D = 36 - 36 = 0$ gives equal roots (3 and 3): A true; R is false. |
| **49** | **(c)** A is true but R is false. | $a_{10} = 2 + 9 \times 3 = 29$, so A is true; the correct formula is $a + (n - 1)d$, so R is false. |
| **50** | **(c)** A is true but R is false. | $AC^2 = 36 + 64 = 100$, so $AC = 10$: A true; the theorem holds for every right-angled triangle, so R is false. |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

**Full Syllabus (Chapters 1–14)** | Class 10 Maths | Maximum Marks: 50

### MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 14 | Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$ |
| 2 | **(c)** 32 | $(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$ |
| 3 | **(d)** 59 | $a_n = \frac{2S_n}{n} - a$ |
| 4 | **(b)** 30 | Find $b$ by Pythagoras |
| 5 | **(b)** 44 | Twice the area |
| 6 | **(b)** $-\frac{119}{169}$ | $\cos^2\theta - \sin^2\theta$ |
| 7 | **(c)** $3+3\sqrt{3}$ m | $h(\cot 30^\circ - \cot 45^\circ) = 6$ gives $h = 6 \div (\sqrt{3}-1)$; Height $= 3+3\sqrt{3}$ m (distances $9+3\sqrt{3}$ m and $3+3\sqrt{3}$ m) |
| 8 | **(d)** 26.88 cm | $AM = \frac{14\times 48}{50} = 13.44$ cm; $AB = 2AM = 26.88$ cm |
| 9 | **(b)** 67.82 cm² | Sector $= 115.50$ cm²; triangle with sides $r$, $r$ and angle $120^\circ$ has area $\frac{1}{2}r^2\sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 47.68$ cm²; Segment $= 115.50 - 47.68 = 67.82$ cm² |
| 10 | **(b)** 643.5 cm³ | $\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3}\times\frac{22}{7}(9^3 - 7.5^3)$; Volume $= 643.5$ cm³ |
| 11 | **(c)** 21 | $25 = 20 + \frac{50-(20+x)}{18}\times 10$ gives $x = 21$; $x = 21$; then $y = 100 - 75 - x = 4$ |
| 12 | **(c)** $\frac{1}{6}$ | Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$ |
| 13 | **(d)** 24 | HCF of the differences from the remainder |
| 14 | **(c)** 56 | $\alpha\beta(\alpha+\beta)$ |
| 15 | **(c)** 7 | Solve for $x$ and $y$ |
| 16 | **(d)** 8 | $D = 0$ gives $k^2 = 4m$ |
| 17 | **(c)** 54 | $a_n = S_n - S_{n-1}$ |
| 18 | **(d)** $\frac{60}{13}$ | $\frac{1}{2}ab = \frac{1}{2}ch$ |
| 19 | **(c)** 4 | Equate squared distances |
| 20 | **(a)** $\frac{120}{169}$ | $2\cdot\frac{p}{c}\cdot\frac{q}{c}$ |
| 21 | **(b)** 24 m | Height of the top of the flagstaff $= d\tan 60^\circ = 36$ m; Length $= 36 - 12 = 24$ m |
| 22 | **(c)** 22 cm | Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2\times 11 = 22$ cm |
| 23 | **(d)** 497 mm | Circumference $= \pi d = 154$ mm; diameters $= 7\times 49 = 343$ mm; Total $= 497$ mm |
| 24 | **(c)** 35280 cm³ | Sphere $= \frac{4}{3}\pi\left(\frac{42}{2}\right)^3 = 38808$ cm³; cube $= 74088$ cm³; Wasted $= 35280$ cm³ |
| 25 | **(b)** 71 | $\sum f_i = 100$ and $\sum f_i x_i = 7100$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7100}{100} = 71$ |
| 26 | **(b)** $\frac{2}{9}$ | Favourable outcomes: (1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (6,4); number $= 8$; $P = \frac{8}{36} = \frac{2}{9}$ |
| 27 | **(b)** 2 | $18 = 2\times 3^2$; multiply by the primes that occur to an odd power |
| 28 | **(d)** 7 | $\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$ |
| 29 | **(b)** 6 | Intercepts 4 and 3; area $= \frac{1}{2}\times 4\times 3$ |
| 30 | **(b)** 2 | Substitute the root |
| 31 | **(b)** 6 | $d = 2p$ |
| 32 | **(d)** 18 | Sides in ratio $a : b$ |
| 33 | **(a)** 8 | $x = 0$: $mc + na = 0$ |
| 34 | **(c)** $\frac{1}{5}$ | $(\sec+\tan)(\sec-\tan) = 1$ |
| 35 | **(d)** 4 minutes | Distance covered in 8 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it; Time left $= 8 \div 2 = 4$ minutes |
| 36 | **(d)** 43 cm | $12x + 7 = 5x + 28$ gives $x = 3$; $QA = 12(3) + 7 = 43$ cm |

| | | |
|---|---|---|
| **37** | **(d)** 2772 m² | Area $= \frac{180}{360} \times \frac{22}{7} \times 42^2$; Area $= 2772$ m$^2$ |
| **38** | **(b)** 19404 cm³ | Cylinder $= \pi r^2(2r) = 58212$ cm$^3$; sphere $= \frac{4}{3}\pi r^3 = 38808$ cm$^3$; Unoccupied $= 19404$ cm$^3$ |
| **39** | **(c)** 190 | Median class 180–200: $l = 180$, $cf = 39$, $f = 22$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 180 + \frac{50-39}{22} \times 20 = 190$ |
| **40** | **(c)** 44 | $\frac{x}{11+x} = 4 \times \frac{11}{11+x}$ gives $x = 4 \times 11$; $x = 44$ |
| **41** | **(c)** A is true but R is false. | $\left(\frac{2+8}{2}, \frac{-3+5}{2}\right) = (5, 1)$: A true; the formula uses sums, not differences, so R is false. |
| **42** | **(c)** A is true but R is false. | $\tan 45^\circ = \frac{1/\sqrt{2}}{1/\sqrt{2}} = 1$: A true; $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, so R is false. |
| **43** | **(c)** A is true but R is false. | $\tan 60^\circ = \dfrac{10\sqrt{3}}{10} = \sqrt{3}$ so A is true; R is false because $\tan 60^\circ = \sqrt{3}$. |
| **44** | **(c)** A is true but R is false. | A is true; a tangent meets the circle at exactly one point, so R is false. |
| **45** | **(c)** A is true but R is false. | Arc $= \frac{1}{6} \times 2 \times \frac{22}{7} \times 21 = 22$ cm: A true; arc length is $\dfrac{\theta}{360^\circ} \times 2\pi r$, so R is false. |
| **46** | **(c)** A is true but R is false. | $\frac{1}{3}\pi r^2 h$ against $\pi r^2 h$: A is true; the cone volume is $\frac{1}{3}\pi r^2 h$, so R is false. |
| **47** | **(c)** A is true but R is false. | A is the correct rule; the class with the highest frequency is the *modal* class, so R is false. |
| **48** | **(c)** A is true but R is false. | Primes are 2, 3, 5, so $P = \frac{3}{6} = \frac{1}{2}$: A true; probability lies between 0 and 1, so R is false. |
| **49** | **(c)** A is true but R is false. | $\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$ is rational, so A is true; R is false (this is a counter-example). |
| **50** | **(d)** A is false but R is true. | $x^2 + 1 \geq 1 > 0$, so its graph never meets the x-axis and it has no real zero: A is false; R is true. |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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